

FROM MEMBRANE DISRUPTION TO RECEPTOR ACTIVATION - MECHANISTIC INSIGHTS INTO PSM α 3 CYTOTOXICITY -

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The virulence of *Staphylococcus aureus*, a multi-drug-resistant pathogen, is tightly linked to the secretion of phenol-soluble modulins (PSMs), notably PSM α 3 which self-assembles into unique cross- α amyloid-like fibrils¹. Here, we investigate the molecular mechanisms underlying the concentration-dependent cytotoxic activities of PSM α 3.

At micromolar concentrations, PSM α 3 directly interacts with lipid membranes, leading to membrane disruption and cell lysis. Combining atomic force microscopy, infrared spectroscopy, and nuclear magnetic resonance, we identified intermediate species as the primary cytotoxic entities, and demonstrated that membrane lipid composition modulates their activity². Notably, PSM α 3 induced-depletion is enhanced by membrane rigidity, *via* cholesterol content or phase separation. At nanomolar concentrations, PSM α 3 activates the G-protein-coupled receptor FPR2, triggering pro-inflammatory signaling and neutrophil recruitment³. Using Förster resonance energy transfer, and confocal microscopy, we demonstrate that FPR2 activation involves the G α i signaling pathway and results in inhibition of cAMP production. Finally, we further reveal that N-terminal capping of PSM α 3, beyond influencing its supramolecular assemblies, critically regulates both its lipid and receptor interactions, likely *via* distinct hydrophobic contributions with acyl chains and binding pocket respectively. Overall, our results provide new mechanistic insights into PSM α 3-mediated cytotoxicity, both *via* lipid and receptor interactions, in the context of *S. aureus* pathogenicity.

1 E. Tayeb-Fligelman, et al., *Science*, **355**, 831-833, (2017), [10.1126/science.aaf4901](https://doi.org/10.1126/science.aaf4901)

2 L. Bonnecaze, et al., *Nanoscale Horiz.*, **9**, 1175-1189, (2024), [10.1039/D4NH00088A](https://doi.org/10.1039/D4NH00088A)

3 D. Kretschmer, et al., *Cell Host Microbe*, **7**, 463-473, (2010), [10.1016/j.chom.2010.05.012](https://doi.org/10.1016/j.chom.2010.05.012)